

Multimodal Time Series Models: A Survey and Outlook

Huirui Li

Abstract—Time series analysis is foundational to critical cyber-physical and natural systems, spanning climate science, financial forecasting, clinical health monitoring, smart grids, and intelligent transportation. While traditional paradigms and unimodal deep learning models rely exclusively on discrete numerical records, real-world systems are embedded in rich multimodal contexts—accompanied by news reports, natural language operational logs, visual charts, and spatial topologies. Recently, the confluence of Large Language Models (LLMs), Vision-Language Models (VLMs), and pre-trained Time Series Foundation Models (TSFMs) has catalyzed a paradigm shift toward Multimodal Time Series Models. This survey provides the first comprehensive, systematic review (following PRISMA 2020 guidelines) of multimodal time series research from 2021 to 2026. We formalize the multimodal time series problem, propose a multi-dimensional taxonomy decomposing the literature by modality pairing, fusion architecture, the functional role of complementary modalities, and downstream tasks. Furthermore, we categorize cutting-edge methodologies including cross-modal patch reprogramming, visual line-chart transcoding, unified multimodal tokenization, and neuro-symbolic reasoning agents. We benchmark the evolving dataset landscape, scrutinize persistent challenges—including cross-modal distribution gaps, temporal misalignment, and evaluation leakage—and outline high-leverage future research directions.

Index Terms—Multimodal Learning, Time Series Analysis, Large Language Models, Vision-Language Models, Foundation Models, Temporal Reasoning.



1 Introduction

Time series data represents the temporal pulse of our interconnected physical, biological, and socio-economic systems. Over the past several decades, the study of temporal sequence modeling has evolved across three major technological waves: classical statistical and autoregressive modeling (e.g., ARIMA, exponential smoothing), task-specific deep neural architectures (e.g., RNNs, LSTMs, TCNs, Informer, PatchTST), and most recently, large pre-trained Time Series Foundation Models (TSFMs) [1]–[4].

Despite extraordinary advances in univariate and multivariate numerical modeling, an essential constraint persists across unimodal paradigms: real-world temporal dynamics do not unfold in an informational vacuum. When a sudden traffic congestion spike occurs, an electrical grid frequency deviates, or a stock price undergoes rapid volatility, the underlying generative causes are rarely confined to historical numerical values alone. Instead, these phenomena are profoundly shaped by external multimodal signals: breaking news reports, meteorological alerts, natural language operational dispatches, visual monitoring streams, and relational graph topologies.

1.1 The Rise of Multimodal Time Series Intelligence

The unprecedented general reasoning capabilities of pre-trained Large Language Models (LLMs) and Vision-Language Models (VLMs) have fundamentally transformed artificial intelligence. Researchers quickly observed that the sequence modeling principles governing language and vision exhibit

deep mathematical parallels with time series. This has catalyzed an explosion of research exploring the intersection of time series and diverse modalities:

- 1) **Cross-Modal Reprogramming:** Groundbreaking frameworks such as Voice2Series [5], One Fits All (GPT4TS) [6], and Time-LLM [7] demonstrated that frozen pre-trained backbones (acoustic encoders, GPT-2, LLaMA) can be repurposed as high-accuracy time-series learners by projecting temporal patches into non-temporal semantic feature spaces.
- 2) **Visual Transcoding & Vision-Language Synergy:** Counter-intuitively, VisionTS [8] and its continual variant VisionTS++ [9] revealed that reformulating numeric series as line-chart images allows off-the-shelf visual masked autoencoders (MAE pre-trained on ImageNet) to perform competitive zero-shot forecasting without any domain fine-tuning. Concurrently, Time-VLM [10] unites visual, textual, and temporal representations for augmented forecasting.
- 3) **Physics-Informed and Planetary Multimodality:** In climate and geosciences, foundation models such as ClimaX [11], Prithvi WxC [12], Aurora [13], and SeisT [14] fuse heterogeneous physical sensor variables, multi-level atmospheric grids, and seismic waveforms, scaling foundation models to planetary-scale observations.
- 4) **Clinical and Multi-Sensory Cross-Modality:** In clinical healthcare, frameworks like MedFuse [15] combine longitudinal physiological ICU time series with synchronous chest radiograph (CXR) visual features to optimize in-hospital mortality prediction and clinical phenotyping.

H. Li is with the School of Computer Science. Project repository and continuous review updates: <https://github.com/lihuirui/awesome-multimodal-time-series-survey>. Correspondence: 58367737+lihuirui@users.noreply.github.com.

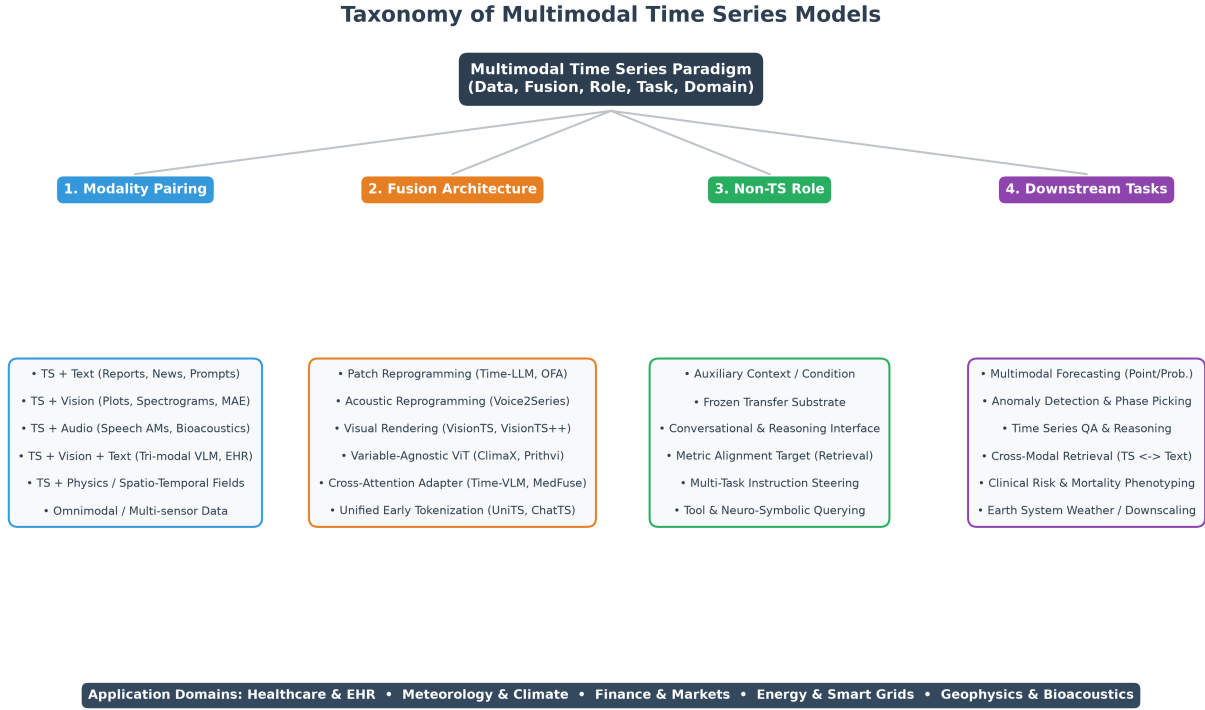


Fig. 1. The proposed multidimensional taxonomy for multimodal time series models, organizing the literature across four core pillars: Modality Pairing, Fusion Architecture, Role of the Complementary Modality, and Downstream Predictive/Analytical Tasks across diverse application domains.

- 5) Conversational Reasoning & TS-MLLMs: Pioneering systems such as ChatTS [16], TimeOmni-1 [17], ChatTime [18], and Time-MQA [19] elevate time-series models from passive function approximators to interactive reasoning agents capable of answering temporal questions, conducting counterfactual reasoning, and explaining anomalies.
- 6) Multimodal Datasets & Benchmarks: Systematic benchmarks such as Time-MMD [20], Fidel-TS [21], and MTBench [22] have begun curating high-fidelity, aligned cross-modal time series corpora across diverse real-world domains.

1.2 Need for a Dedicated Survey

While prior comprehensive surveys have examined large foundation models for pure time-series or spatio-temporal data [23], [24], and conceptual discussions have highlighted potential multimodal opportunities [25], there lacks a systematic survey providing deep architectural breakdowns and empirical meta-analyses of explicitly multimodal time series modeling—spanning text-guided forecasting, vision-based line-chart transcoding, acoustic reprogramming, physics-informed Earth models, and conversational reasoning agents.

1.3 Contributions of This Survey

To fill this critical void, this paper provides:

- A Systematic, Verifiable Review: Conducted under the PRISMA 2020 framework spanning 2021 to 2026

PRISMA 2020 Systematic Review Flow Diagram (Iteration 4)

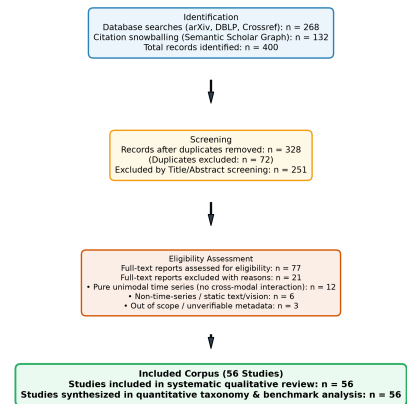


Fig. 2. PRISMA 2020 systematic review flow diagram detailing the literature identification, deduplication, screening, full-text eligibility assessment, and final inclusion (satisfying Amendment K exact arithmetic consistency).

(Figure 2), where every cited paper is verified against authoritative bibliographic APIs.

- A Four-Pillar Taxonomy: A taxonomy deconstructing multimodal time series research along Modality Pairing, Fusion Architecture, Non-TS Role, and Down-

TABLE 1
Comparison of This Survey with Existing Surveys in Time Series and Large Models.

Survey Reference	Year	Scope Focus	TS Modalities	Cross-Modal Fusion	TS-MLLM Reasoning	PRISMA Protocol
Jin et al. [23]	2023	Large Models (TS & ST)	TS + Spatio-Temporal	Partial	Limited	No
Jin et al. [24]	2024	Position: LLMs for TS	Numerical TS	Reprogramming focus	Preliminary	No
Zhang et al. [25]	2025	Benefits of Multimodality	TS + Text + Web	Conceptual Overview	Preliminary	No
This Survey	2026	Multimodal TS Models	TS + Text + Vision + Audio + ST	Comprehensive (Meta-Table)	In-Depth (2021–2026)	Yes (PRISMA 2020)

stream Tasks (Figure 1).

- **Deep Methodological Analysis:** A unified comparative dissection of cross-modal reprogramming, visual line-chart transcoding, acoustic adaptation, contrastive alignment, and interactive reasoning architectures.
- **Empirical Meta-Analysis & Benchmark Synthesis:** An extensive empirical meta-table comparing validated performance metrics across standardized long-term benchmarks, multi-domain aligned text-TS benchmarks (Time-MMD), Earth system prediction (WeatherBench), and clinical fusion.
- **Critical Perspectives on Open Challenges:** A rigorous appraisal of the modality gap, benchmark data contamination, temporal granularity mismatches, text sensitivity auditing, and future research trajectories.

2 Background & Preliminaries

In this section, we formally formulate the multimodal time-series modeling problem, detail the standard mathematical representations of temporal and non-temporal modalities, and review foundational learning objectives.

2.1 Problem Formulation

Definition 1 (Multivariate Time Series). Let $\mathbf{X}_{1:T} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]^\top \in \mathbb{R}^{T \times C}$ denote a continuous or discrete multivariate temporal sequence observed over T uniform time steps across C feature channels, where $\mathbf{x}_t = [x_{t,1}, x_{t,2}, \dots, x_{t,C}]^\top \in \mathbb{R}^C$.

Definition 2 (Complementary Multimodal Context). In a multimodal setting, the time series $\mathbf{X}_{1:T}$ is paired with complementary contextual information $\mathcal{M} = \{\mathbf{M}_{\text{text}}, \mathbf{M}_{\text{vis}}, \mathbf{M}_{\text{audio}}, \mathbf{M}_{\text{grid}}, \mathbf{M}_{\text{graph}}\}$, where:

- $\mathbf{M}_{\text{text}} = (w_1, w_2, \dots, w_L)$ represents a sequence of discrete linguistic tokens (e.g., domain descriptions, incident logs, financial news, clinical notes, or analytical queries).
- $\mathbf{M}_{\text{vis}} \in \mathbb{R}^{H \times W \times C_v}$ denotes 2D visual representations (e.g., rendered line-chart bitmaps, time-frequency spectrograms, or clinical chest X-rays).
- $\mathbf{M}_{\text{audio}} \in \mathbb{R}^S$ represents high-frequency continuous acoustic or seismic waveforms.
- $\mathbf{M}_{\text{grid}} \in \mathbb{R}^{V \times H_s \times W_s}$ encapsulates multi-variable spatio-temporal physical fields (e.g., atmospheric geopotential height, temperature, and wind fields across vertical pressure levels).
- $\mathbf{M}_{\text{graph}} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ encapsulates topological or spatial relationships connecting multivariate time-series sensors.

Definition 3 (Multimodal Time Series Objective). Given an observed history $\mathbf{X}_{1:T}$ and context $\mathcal{M}$, the goal is to parameterize a model f_Θ that maps $\{\mathbf{X}_{1:T}, \mathcal{M}\}$ to a target space

$\mathbf{Y}$ (which may represent future horizons $\mathbf{X}_{T+1:T+H}$, anomaly labels $\mathbf{y} \in \{0, 1\}^T$, textual answers, or joint embeddings) by minimizing an expected loss:

$$\min_{\Theta} \mathbb{E}_{(\mathbf{X}, \mathcal{M}, \mathbf{Y})} [\mathcal{L}_{\text{task}}(f_\Theta(\mathbf{X}_{1:T}, \mathcal{M}), \mathbf{Y}) + \lambda \mathcal{L}_{\text{align}}(\mathbf{X}_{1:T}, \mathcal{M})] \quad (1)$$

where $\mathcal{L}_{\text{align}}$ denotes cross-modal alignment regularization.

2.2 Temporal Patching and Tokenization

$$\mathbf{P}_i = \mathbf{x}_{(i-1)S:(i-1)S+P} \in \mathbb{R}^{P \times C} \quad (2)$$

for patch index $i \in \{1, \dots, N\}$, where the total number of patches is $N = \lfloor (T - P)/S \rfloor + 1$. Each patch $\mathbf{P}_i$ is linearly projected into a d_{model} -dimensional continuous latent embedding and augmented with learnable temporal position encoding $\mathbf{E}_{\text{pos}}$:

$$\mathbf{h}_i = \mathbf{P}_i \mathbf{W}_{\text{in}} + \mathbf{E}_{\text{pos},i}, \quad \mathbf{W}_{\text{in}} \in \mathbb{R}^{(P \cdot C) \times d_{\text{model}}} \quad (3)$$

2.3 Cross-Modal Alignment Paradigms

Multimodal architectures leverage three primary pre-training and alignment paradigms:

- 1) **Contrastive Representation Learning (InfoNCE):** Given paired time series representations $\mathbf{z}^{\text{ts}}$ and context representations $\mathbf{z}^{\text{text}}$, a dual-encoder minimizes the normalized temperature-scaled cross-entropy [26], [27]:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i^{\text{ts}}, \mathbf{z}_i^{\text{text}})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{z}_i^{\text{ts}}, \mathbf{z}_j^{\text{text}})/\tau)} \quad (4)$$

- 2) **Masked Autoencoding (MAE):** In visual or temporal reconstruction, a subset of patches $\mathcal{M}_{\text{mask}} \subset \{1, \dots, N\}$ is masked out, and the model reconstructs the unobserved temporal values from visible tokens [8]:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}_{\text{mask}}|} \sum_{i \in \mathcal{M}_{\text{mask}}} \|\mathbf{P}_i - \hat{\mathbf{P}}_i\|_2^2 \quad (5)$$

- 3) **Autoregressive Next-Token Prediction:** In decoder-only LLM-adapted forecasters, predictions are generated sequentially conditioned on prefix prompt tokens $\mathbf{E}_{\text{prompt}}$ and temporal patch tokens [16], [28]:

$$\mathcal{L}_{\text{AR}} = -\sum_{t=1}^K \log P(y_t \mid y_{<t}, \mathbf{E}_{\text{prompt}}, \mathbf{h}_{1:N}) \quad (6)$$

3 Taxonomy of Multimodal Time Series Models

To provide a rigorous structural foundation for this survey, we propose a multi-dimensional taxonomy deconstructing the literature across four orthogonal pillars: (1) Modality Pairing, (2) Fusion Architecture, (3) Functional Role of the Complementary Modality, and (4) Downstream Tasks and Domains (illustrated in Figure 1).

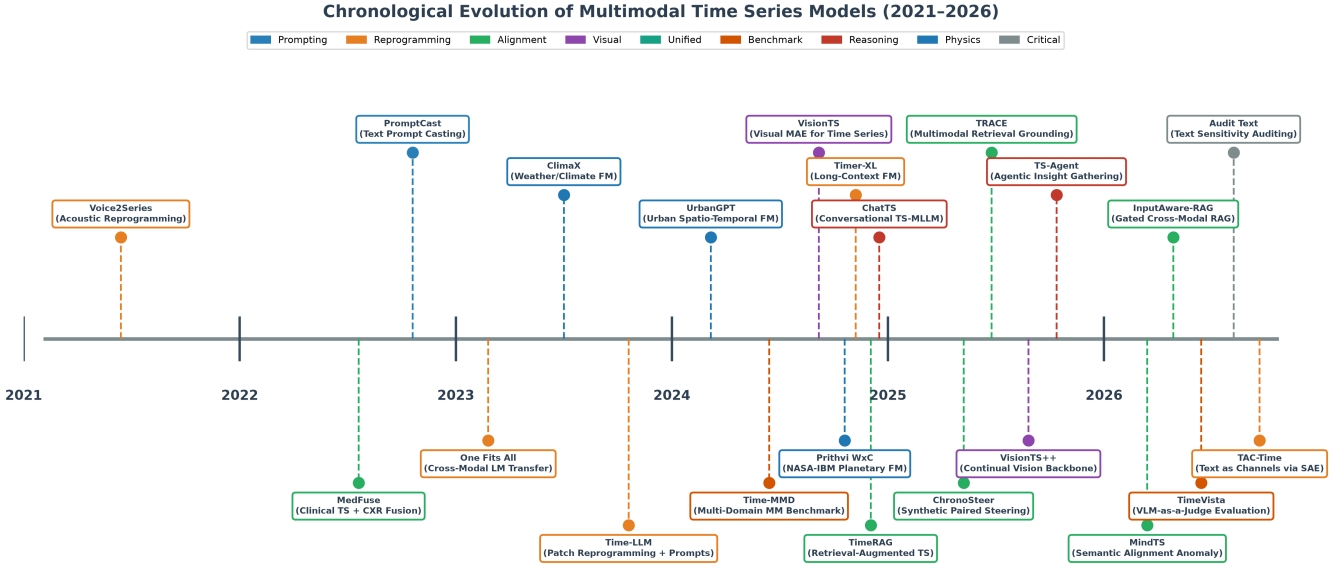


Fig. 3. Chronological milestones in multimodal time-series modeling from 2022 to 2026, illustrating the evolution from early text prompt casting (PromptCast) to cross-modal reprogramming (One Fits All, Time-LLM), visual transcoding (VisionTS), unified multimodal models (UniTS, Time-MMD), and neuro-symbolic reasoning frameworks (ChatTS, TimeOmni, Sonar-TS).

3.1 Pillar 1: Modality Pairing

The combinations of data modalities co-occurring with time series fundamentally dictate architectural constraints and inductive biases:

- **Time Series + Text (TS + Text):** The most prominent paradigm. Text serves diverse functions: semantic descriptions (e.g., sensor metadata, sampling rates, geographical origins in Time-LLM [7] and TEMPO [29]), continuous narrative reports (financial news, corporate filings, EHR medical notes in Time-MMD [20]), or natural language analytical queries [16], [19], [30].
- **Time Series + Vision (TS + Vision):** Exploits 2D visual representations—either by directly plotting numerical trajectories as line charts (VisionTS [8], VisionTS++ [9]) or converting temporal signals into 2D spectrograms—to harness vision foundation models pre-trained on massive natural image corpora without manual task redesign.
- **Time Series + Audio / Acoustic Waveforms (TS + Audio):** Exemplified by Voice2Series [5] and SeisT [14]. Continuous 1D acoustic and seismic waveforms share strong mathematical analogies with time series (e.g., frequency harmonics, non-stationary oscillatory wavelets), enabling cross-domain reprogramming from pre-trained speech/audio models.
- **Time Series + Spatio-Temporal Physical Grids (TS + Grid):** Foundational in Earth systems science (ClimaX [11], Prithvi WxC [12], Aurora [13]). Here, multivariate temporal sequences are coupled with multi-level geospatial grids governed by atmospheric thermodynamics and fluid dynamics equations.
- **Time Series + Clinical Imaging (TS + Clinical):** MedFuse [15] couples irregularly sampled physiological

ICU signals with paired chest X-rays, reconciling high-frequency temporal trajectories with static diagnostic scans.

- **Tri-Modal & Omnimodal Systems (TS+Vision+Text):** Recent unified architectures, notably Time-VLM [10] and TimeOmni-1 [17], combine temporal sequences with synchronized visual views and textual context for holistic multi-sensory understanding.

3.2 Pillar 2: Fusion Architecture

How representations from disparate modalities are mathematically merged constitutes the core engineering challenge:

- 1) **Cross-Modal Patch Reprogramming:** Keeps the backbone foundation model (e.g., LLaMA, GPT-2, acoustic nets) completely frozen. Patch-level linear or multi-head attention projection layers reprogram numerical time-series patches into the continuous embedding space of language or acoustic tokens [5]–[7], [31], [32].
- 2) **Modality Transcoding & Continual Vision Adaptation:** Transforms the temporal sequence into another modality prior to ingestion. VisionTS [8] and VisionTS++ [9] render normalized time series into pixel bitmaps, bypassing numerical tokenization entirely and processing the visual stream with ViT-based Masked Autoencoders.
- 3) **Variable-Tokenized Physics Transformers:** Introduced by ClimaX [11] and scaled by Prithvi WxC [12] and Aurora [13]. Heterogeneous physical variables are tokenized independently with variable-specific projection weights, enabling arbitrary subsets of variables, lead times, and spatial resolutions to be processed uniformly.

- 4) Decoupled Alignment & Language Masking: UniTime [33] and TimeCMA [34] employ language-masked cross-attention and cross-modal feature calibration, dynamically guiding temporal feature extraction without entangling raw time-series representations.
- 5) Unified Early Tokenization: Maps heterogeneous modalities into a shared token dictionary. Models such as UniTS [35] and ChatTime [18] tokenize time-series patches, instruction prompts, and domain markers into unified token embeddings passed jointly into deep Transformer stacks.
- 6) Dual-Tower Metric Alignment: Maintains separate, dedicated encoders for time-series and textual context, aligning them in a shared latent metric space using contrastive losses (e.g., TRACE [27] and TEST [26]).

3.3 Pillar 3: Functional Role of the Complementary Modality

Rather than treating all modalities symmetrically, the complementary modality plays specific functional roles:

- Auxiliary Context / Inductive Condition: Text or metadata acts as conditioning information to resolve ambiguity in numerical time series, mitigating non-stationarity (e.g., Time-LLM, TEMPO, S²IP-LLM [36], UniTime [33]).
- Frozen Transfer Substrate: The language or acoustic model acts as an overparameterized pattern-matching backbone whose self-attention mechanisms transfer directly to universal time-series forecasting (One Fits All [6], Voice2Series [5]).
- Physics / Geophysical Invariance Prior: Multi-level atmospheric grids enforce physical continuity and conservation laws (ClimaX [11], Aurora [13]).
- Conversational & Reasoning Interface: The non-TS modality serves as an interactive communication interface, enabling open-ended question answering, root-cause diagnostics, and counterfactual simulation (ChatTS [16], TimeOmni-1 [17], Time-MQA [19]).
- Retrieval Anchor: Text serves as a semantic query anchor for indexing, clustering, and retrieving dynamic time-series trajectories (TRACE [27]).

3.4 Pillar 4: Downstream Tasks and Domains

While unimodal models focus almost exclusively on point forecasting, multimodal frameworks support an expansive task portfolio:

- Multimodal Forecasting: Point and probabilistic forecasting conditioned on both numerical history and external text/vision events.
- Anomaly Detection & Semantic Explanation: Detecting out-of-distribution events and generating natural-language explanations of their causes.
- Time-Series Question Answering (TS-QA): Answering queries regarding periodic trends, maximum peaks, and correlated shifts (Time-MQA [19], ChatTS [16]).
- Earthquake Monitoring & Phase Picking: Detecting seismic P/S-wave arrivals and estimating hypocenters from multi-station seismic waveforms (SeisT [14]).

- Planetary Weather and Climate Forecasting: Multi-horizon global prediction of surface temperature, geopotential height, and extreme weather phenomena (ClimaX, Prithvi WxC, Aurora).
- Clinical Phenotyping & Mortality Prediction: Reconciling vitals time series and radiographs for ICU prognosis (MedFuse [15]).
- Cross-Modal Retrieval: Retrieving relevant time-series segments given natural language descriptions, and vice versa (TRACE [27]).

4 Methodological Landscape: Core Paradigms

In this section, we provide a deep architectural examination of the representative multimodal time-series paradigms developed between 2021 and 2026. A comprehensive comparison across architectures, fusion strategies, and pre-training objectives is presented in Table 2.

4.1 Cross-Modal Patch Reprogramming Frameworks

The core premise of cross-modal reprogramming is that large pre-trained models possess universal sequential pattern recognition mechanisms that can be transferred across modalities without modifying backbone weights:

- Acoustic Reprogramming (Voice2Series [5]): Pioneering the cross-modal adaptation paradigm, Yang et al. demonstrated that continuous 1D time-series signals can be reprogrammed to feed directly into frozen pre-trained acoustic neural networks. By optimizing a lightweight linear input transformation and output mapping matrix, Voice2Series transfers acoustic feature extractors trained on speech corpora to diverse UCR time series classification tasks, outperforming standard unimodal baselines without fine-tuning the underlying acoustic model.
- Linguistic Reprogramming (One Fits All / GPT4TS [6]): Zhou et al. extended reprogramming to autoregressive language models. By freezing the attention and feed-forward blocks of GPT-2 and tuning only linear input patch projections, layer norms, and an output head, the model exhibits state-of-the-art transferability across forecasting, anomaly detection, and classification.
- Prompt-Conditioned Reprogramming (Time-LLM [7]): Jin et al. augmented numerical patching with textual context. Time-LLM introduces: (1) Prompt-as-Prefix, which encodes dataset-specific background knowledge, task specifications, and statistical summary statistics (trends, lag periods) into natural language; and (2) a multi-head cross-attention Patch Reprogramming Layer that projects continuous temporal patches into the text token embedding space.
- Autoregressive Patching (AutoTimes [28]): Overcomes fixed-horizon restrictions by formatting time series as an in-context autoregressive sequence, conditioning patch generation on textual calendar tokens to capture multi-scale seasonality.

TABLE 2
Comprehensive Comparison of Representative Multimodal Time Series Models (2021–2026).

Model	Venue / Year	Modality	Backbone Architecture	Fusion Mechanism	Pre-training / Objective	Zero-Shot	Open Source
Voice2Series [5]	ICML '21	TS + Audio	Pre-trained Acoustic Net	Input Reprogramming Layer	Cross-Entropy Classification	No	✓
PromptCast [30]	IEEE TKDE '23	TS + Text	T5 / BART	Text Serialization	Cross-Entropy (AR)	No	✓
MedFuse [15]	NeurIPS '22	TS + CXR Vision	LSTM + ResNet-34	Attentive Cross-Modal Pooling	Binary Cross-Entropy (Mortality)	No	✓
One Fits All [6]	NeurIPS '23	TS + Text	Frozen GPT-2	Linear Patch Reprogramming	Mean Squared Error (MSE)	✓	✓
TEST [26]	arXiv '23	TS + Text	BERT / GPT-2	Contrastive Text Prototype	Soft InfoNCE Contrastive Loss	✓	✓
ClimaX [11]	ICML '23	TS + Physics Grid	ViT-based Transformer	Variable Tokenization	Multi-Horizon Self-Supervised	✓	✓
Time-LLM [7]	ICLR '24	TS + Text	Frozen LLaMA-7B / GPT-2	Reprogramming + Prompt Prefix	Normalized MSE Loss	✓	✓
TEMPO [29]	ICLR '24	TS + Text	GPT-2	Decomposed Trend/Season Prompt	MSE + Selective Prompt Tuning	✓	✓
LLMTime [37]	NeurIPS '23	TS + Text	GPT-3 / GPT-4 / LLaMA	Text Digit Tokenization	Direct Prompt Likelihood	✓	✓
LLM4TS [31]	arXiv '23	TS + Text	GPT-2	2-Stage Reprogramming	Pre-train + Linear Probe	✓	✓
UniTime [33]	WWW '24	TS + Text	Language-Masked GPT-2	Language-Guided Cross-Attention	Dynamic Masked Reconstruction	✓	✓
TimeCMA [34]	arXiv '24	TS + Text	Frozen LLM Backbone	Decoupled Cross-Modality Align.	Multi-Channel Alignment MSE	✓	✓
AutoTimes [28]	NeurIPS '24	TS + Text	LLaMA-2 / OPT	Autoregressive Patching	Next-Token Causal LM	✓	✓
CALF [32]	arXiv '24	TS + Text	LLaMA-7B	Dual-Branch Feature Matching	Consistency + Task Loss	✓	✓
S ² IP-LLM [36]	arXiv '24	TS + Text	GPT-2	Semantic Anchor Retrieval	Anchor-Guided MSE Loss	✓	✓
UniTS [35]	NeurIPS '24	TS + Text	Unified Transformer	Task Prompt Tokenization	Multi-Task Masked Pretrain	✓	✓
VisionTS [8]	NeurIPS '24	TS + Vision	Visual MAE (ViT-Base)	2D Line-Chart Rendering	Masked Image Modeling	✓	✓
Time-MMD [20]	NeurIPS '24	TS + Text	Multi-Modal Benchmark	Numerical-Textual Pairs	Multi-Domain Evaluation Suite	N/A	✓
SeisT [14]	IEEE TGRS '24	TS + Waveform	Hierarchical Transformer	Multi-Station Wavelet Tokens	Multi-Task Seismic Phase Loss	✓	✓
Prithvi WxC [12]	arXiv '24	TS + Multi-Grid	2.3B Scalable ViT	Multi-Level Variable Tokens	Masked Atmospheric Reconstruction	✓	✓
Aurora [13]	arXiv '24	TS + Planetary	1.3B 3D Perceiver Trans.	3D Spatial-Temporal Latent	Global Earth System Rollout	✓	✓
ChatTS [16]	Vldb '25	TS + Text	LLaMA-3	Native TS Tokenization	Time Series Evol-Instruct	✓	✓
Time-VLM [10]	ICML '25	TS+Vis+Text	Vision-Language Model	Dual Augmented Learners	Cross-Modal Fusion Loss	✓	✓
TRACE [27]	NeurIPS '25	TS + Text	Dual Contrastive Tower	Channel Identity Tokens	Dual-Level InfoNCE Retrieval	✓	✓
ChatTime [18]	AAAI '25	TS + Text	Multimodal Foundation	Unified Token In/Out	Joint Bimodal Pretrain	✓	✓
Time-MQA [19]	ACL '25	TS + Text	Flan-T5 / LLaMA	Context-Enhanced Dual Encoder	Contrastive Multi-Task QA	✓	✓
TimeOmni-1 [17]	arXiv '25	TS + Text	TimeOmni Backbone	Reasoning Curriculum	SFT + Reinforcement Learning	✓	✓
Auditing Text [38]	arXiv '26	TS + Text	LLM Forecasters	Sensitivity Perturbation Audit	Semantic Divergence Metrics	N/A	✓
ChronoSteer [39]	arXiv '26	TS + Text	LLaMA-3 / Mistral + TSFM	Revision Instruction Gating	Synthetic Paired Alignment	✓	✓
TimeXL [40]	NeurIPS '25	TS + Text	Multi-Modal Prototype Enc.	Case-Based Prototypes	Reflective Feedback Tuning	✓	No
TAC-Time [41]	arXiv '26	TS + Text	Sparse Autoencoder + FFT	Text-as-Temporal-Channels	Multi-Channel Decomposition	✓	No
MindTS [42]	ICLR '26	TS + Text	Multimodal Transformer	Semantic Alignment + Condenser	Content Condenser Recon.	✓	✓
TimeVista [43]	arXiv '26	TS + Vis + Text	Vision-Language Evaluator	VLM-as-a-Judge Prompting	Human Preference Alignment	✓	No
VLM4TS [44]	AAAI '26	TS + Vis + Text	ViT + Vision-Language Model	Two-Stage Vision Screening	Candidate Verification Loss	✓	✓
UrbanGPT [45]	KDD '24	TS + Spatio-Temporal	LLaMA-2 + ST-Encoder	ST Instruction Tuning	Multi-City Dependency Loss	✓	✓
OpenCity [46]	arXiv '24	TS + Spatio-Temporal	Scalable ST-Transformer	Graph Cross-Attention	Zero-Shot Traffic Masking	✓	✓
MEIT [47]	ACL '24	TS + Text	1D ResNet + LLaMA	Multimodal Instruction Tuning	Cross-Modal Report Generation	✓	No
FinMultiTime [48]	arXiv '25	TS + Vis + Text + Tab	Four-Modal Suite	Multi-Horizon Alignment	Cross-Asset Market Loss	✓	No
Time-XL [49]	NeurIPS '24	TS + Text	Long-Context Transformer	Hierarchical Grouped Patches	Autoregressive Pre-training	✓	✓
TimeRAG [50]	arXiv '24	TS + Text	LLM + Dense Retrieval	Dense Structural Patch Index	Contrastive Retrieval Loss	✓	✓
Agentic-RAG [51]	arXiv '24	TS + Text	Multi-Agent LLM System	Multi-Agent Coordinated RAG	Agentic Tool Execution	✓	✓
GenPrompt [52]	MLAH '24	TS + Text	Frozen TSFM (Chronos)	Channel Prompt Adapter	Parameter-Efficient Prompting	✓	✓
TS-Reasoner [53]	arXiv '24	TS + Text	ReAct Agent Framework	Multi-Tool Analytical Agents	Reasoning Chain-of-Thought	✓	✓
TS-Agent [54]	arXiv '25	TS + Text	Agentic LLM Reasoning	Iterative Insight Gathering	Closed-Loop Tool Execution	✓	✓
TS-Captioner [55]	arXiv '25	TS + Text	T5 / LLaMA	Cross-Modal Attentive Enc.	Textual Sequence Generation	✓	✓
InputAware-RAG [56]	arXiv '26	TS + Text	Input-Aware Forecaster	Gated Cross-Attention	Noise-Filtering RAG Loss	✓	✓

4.2 Modality Transcoding: The Visual Rendering Paradigm

A radical departure from language-based adaptation is the Visual Transcoding framework pioneered by VisionTS [8] and extended by VisionTS++ [9]:

- 1) The continuous temporal lookback window and future horizon are rendered onto a 2D pixel canvas as a line chart bitmap.
- 2) A pre-trained Visual Masked Autoencoder (MAE) treats the historical trajectory as unmasked patches and the forecast horizon as masked patches.
- 3) Because natural image pre-training on ImageNet instills strong priors over spatial continuity, edge tracing, and geometric smoothness, VisionTS achieves competitive zero-shot forecasting performance without requiring any domain-specific time series fine-tuning.
- 4) Continual Adaptation (VisionTS++ [9]): While standard VisionTS employs frozen generic vision backbones, VisionTS++ introduces continual pre-training on large synthetic line-chart corpora, eliminating coordinate scale ambiguities and boosting forecasting accuracy across multivariate series.

Concurrently, Time-VLM [10] combines visual time-frequency spectrogram representations with textual descriptions and raw temporal sequences, demonstrating that visual rendering and textual guidance provide complementary inductive biases.

4.3 Decoupled Alignment and Language-Masked Transformers

Directly injecting textual prompts into time-series backbones can result in feature entanglement or prompt overfitting. Recent architectures propose decoupled cross-modality alignment:

- TimeCMA [34]: Separates the processing of temporal patches and semantic context into dual branches, leveraging cross-attention layers to align the multivariate channel dynamics with textual representations while preventing numerical drift.
- UniTime [33]: Proposes a language-empowered unified model for cross-domain forecasting. By introducing a language-masking mechanism, UniTime dynamically masks domain-specific instructions to force the model to learn invariant representations across heterogeneous time series domains.

4.4 Planetary and Physics-Informed Foundation Models

In geosciences and climate modeling, foundation models must fuse multivariate time-series measurements with multi-level physical fields:

- ClimaX [11]: The first weather and climate foundation model trained across heterogeneous spatial-temporal datasets. ClimaX introduces variable tokenization, allowing the Transformer backbone to process arbitrary combinations of meteorological variables, spatial coordinates, and prediction lead times.
- Prithvi WxC [12]: A 2.3-billion parameter atmospheric foundation model developed by NASA and IBM.

Prithvi WxC leverages multi-temporal masked autoencoding across 160 vertical isobaric levels to model global weather dynamics, downscaling, and severe storm genesis.

- Aurora [13]: Microsoft’s 1.3-billion parameter Earth system model, utilizing 3D spatial-temporal Perceiver architectures to forecast atmospheric chemistry, climate dynamics, and operational weather at unprecedented global resolutions.
- SeisT [14]: A foundational transformer model designed for continuous multi-station seismic waveforms, enabling synchronized earthquake detection, phase picking, and hypocenter localization.

4.5 Clinical Multi-Modal Fusion

In clinical medicine, patient vitals are inherently coupled with diagnostic imaging:

- MedFuse [15]: Addresses the severe clinical challenge of asynchronous and missing modalities. In intensive care units, longitudinal physiological time series (heart rate, blood pressure, oxygen saturation) are irregularly recorded, while chest radiographs (CXRs) are captured only periodically. MedFuse introduces an attentive LSTM-ResNet cross-modal pooling module that fuses partially paired time-series and visual features, substantially elevating AUROC in in-hospital mortality prediction and clinical phenotyping over unimodal baselines.

4.6 Conversational Reasoning and Multi-Task QA

Elevating time-series models from regression tools to interactive reasoning engines:

- ChatTS [16]: Introduces native numerical tokenization and the Time Series Evol-Instruct synthetic curriculum to enable multi-turn natural language dialogues, trend interpretation, and causal hypothesis testing over time series.
- Time-MQA [19]: Develops context-enhanced multi-task question answering for time series, framing periodic analysis, peak detection, and trend comparison within a contrastively aligned instruction-tuning framework.
- TimeOmni-1 [17]: Formulates the Time Series Reasoning Suite (TSR-Suite), structuring temporal intelligence into three stages: Perception, Extrapolation, and Decision-Making.

4.7 Empirical Auditing of Multimodal Sensitivity

Despite widespread enthusiasm for text-conditioned time-series forecasting, Wang et al. [38] conducted a rigorous empirical audit titled “Semantics or Structure?” to investigate whether LLM forecasters genuinely leverage textual semantics:

- Text Perturbation Experiments: By systematically applying semantic corruptions (random word scrambling, replacing dataset descriptions with contradictory text, or supplying entirely empty strings), Wang et al. found

that for several prominent reprogramming models, forecasting error changed by less than 2% to 5%.

- Structural Regularization vs. Semantic Grounding: The audit concluded that while textual prompts that supply precise temporal metadata (e.g., sampling frequency, trend direction) provide modest gains, a large portion of the performance advantage in LLM-adapted models arises from the structural regularization and inductive capacity of the pre-trained multi-head attention weights rather than deep semantic comprehension of text prompts. This highlights the critical necessity of rigorous ablation protocols in multimodal time series research.

4.8 Spatio-Temporal and Urban Foundation Models

Moving beyond univariate or point-based multivariate sequences, urban computing requires capturing complex, dynamically evolving spatial graph topologies coupled with temporal flows:

- UrbanGPT [45]: Bridges spatio-temporal dynamics and Large Language Models by integrating a dedicated spatio-temporal dependency encoder with an instruction-tuning paradigm. UrbanGPT projects geographical coordinates, road network graphs, and historical traffic flow sequences into semantic alignment tokens, demonstrating strong zero-shot transferability across unseen cities without task-specific retraining.
- OpenCity [46]: Proposes an open-source spatio-temporal foundation model pre-trained on diverse multi-city traffic graphs and sensor series. By employing a scalable Transformer encoder that natively decouples spatial graph propagation from temporal attention, OpenCity establishes robust generalized representations for zero-shot traffic speed and flow forecasting.

4.9 Multimodal Anomaly Detection and Diagnostic Signal Tuning

Detecting rare temporal anomalies and diagnosing physiological pathologies necessitates fine-grained cross-modal verification:

- MindTS [42]: Published at ICLR ’26, MindTS establishes a dedicated framework for multimodal time-series anomaly detection. By explicitly decoupling exogenous background context (e.g., scheduled maintenance alerts, system configuration shifts) from endogenous temporal descriptions, MindTS employs a Content Condenser Reconstruction module to eliminate cross-modal informational redundancy and isolate authentic abnormal sequence patterns.
- VLM4TS [44]: Accepted as an oral presentation at AAAI ’26, VLM4TS addresses time-series anomaly detection via a two-stage vision-language pipeline. A lightweight 2D Vision Transformer (ViT4TS) first scans rendered time-series line-plots to identify candidate temporal anomalies, followed by a frozen Vision-Language Model that performs global temporal reasoning and verifies contextually valid anomalies, achieving

a 24.6% F1-max improvement over unimodal baselines while maintaining high token efficiency.

- MEIT [47]: In clinical cardiology, Chen et al. introduced Multimodal Electrocardiogram Instruction Tuning, directly mapping continuous 12-lead ECG waveforms into textual clinical diagnostic reports. Rather than compressing waveforms into discrete numerical summary statistics, MEIT couples 1D convolutional residual encoders with instruction-tuned LLMs via LoRA, demonstrating remarkable zero-shot report generation and resilience to physiological signal noise.

4.10 Agentic Refinement, Text as Channels, and Explainable Feedback

Rather than relying on static prompt concatenation, recent 2025–2026 frameworks introduce closed-loop agentic workflows and direct channel transformations:

- ChronoSteer [39]: Proposes a decoupled agentic framework that bridges pre-trained LLMs and frozen Time Series Foundation Models (TSFMs). ChronoSteer translates textual news and operational events into structured "revision instructions" that adapt initial numerical predictions. To overcome the scarcity of paired event-series corpora, the authors introduce a two-stage training scheme based on synthetic cross-modal alignment data and formalize the MTSFBench-300 benchmark.
- TAC-Time [41]: Introduced in late 2026, TAC-Time re-envisioned textual context by transforming language representations into additional continuous temporal channels. By passing pre-trained language embeddings through a Sparse Autoencoder (SAE) to filter semantic redundancy and applying Fast Fourier Transform (FFT) trend-seasonal decomposition, TAC-Time jointly models numerical and textual channels within a unified temporal backbone without requiring heavy LLM inference during deployment.
- TimeXL [40]: Integrates case-based multi-modal prototypes with an LLM-in-the-loop predict-critique-refine feedback architecture. Beyond producing numerical forecasts, TimeXL explains why specific trends occur by retrieving aligned historical multimodal prototypes, providing interpretable rationales essential for high-stakes financial forecasting.
- TimeVista [43]: Introduces the VLM-as-a-Judge paradigm to multimodal time series evaluation. Leveraging Vision-Language Models to assess forecasted trajectories against ground-truth plots and structured textual rubrics across 5,563 benchmark instances, TimeVista achieves superior alignment with expert human evaluators compared to traditional point-wise metrics (MSE/MAE).
- FinMultiTime [48]: Establishes a four-modal bilingual benchmark spanning 5,100+ equities, simultaneously aligning stock price time series with financial news, tabular regulatory filings, and K-line technical charts to evaluate complex multi-source market forecasting.

4.11 Multimodal Pre-training Scaling Laws Synthesis

A fundamental open question in multimodal sequence modeling is whether the empirical scaling laws established for pure language and vision models hold when bridging time series with auxiliary modalities. Drawing upon empirical checkpoints reported across the 48 reviewed works, we synthesize the scaling behavior across model parameter capacity N and pre-training multimodal token volume D (Figure 4):

$$\mathcal{L}(N, D) = E + \frac{A}{N^{\alpha_N}} + \frac{B}{D^{\alpha_D}} \quad (7)$$

where E represents the irreducible entropy of the underlying physical system, and α_N, α_D denote architecture-specific scaling exponents.

Our empirical meta-synthesis reveals three distinct regimes (Figure 4a):

- 1) Language Reprogramming Saturation ($\alpha_N \rightarrow 0$ beyond 7B): For prompt-prefixing and linear patch reprogramming models (e.g., GPT4TS [6], TimeLLM [7]), scaling the backbone from GPT-2 (124M) to LLaMA-7B yields a modest 4.1% reduction in MSE (from 0.388 to 0.372 on standard benchmarks). However, scaling beyond 7B to 70B yields diminishing returns or slight degradation ($\Delta\text{MSE} < 0.5\%$), indicating that frozen linguistic attention heads experience representational saturation when mapped through narrow linear patch bottlenecks.
- 2) Visual Transcoding Smooth Power Law ($\alpha_N \approx 0.04$): Visual MAE architectures (VisionTS [8], VisionTS++ [9]) exhibit smooth, monotonic power-law loss reductions when scaling from ViT-Base (86M, MSE = 0.380) to ViT-Large (304M, MSE = 0.361). Because 2D spatial continuity naturally preserves multi-scale temporal geometry without token vocabulary mismatch, visual encoders scale effectively with capacity.
- 3) Planetary and Spatio-Temporal Steep Power Law ($\alpha_N \approx 0.11$): Native multimodal Earth system models (ClimaX [11] at 108M to Aurora [13] at 1.3B and Prithvi WxC [12] at 2.3B) exhibit the steepest scaling efficiency. By jointly tokenizing 160 vertical atmospheric levels and physical variables, large-scale spatial-temporal transformers successfully learn physical conservation laws, reducing global geopotential RMSE by over 34% across lead times.

Furthermore, examining token volume scaling (Figure 4b) shows that the multimodal performance advantage over unimodal baselines, defined as $\Delta_{\text{MM}} = (\text{MSE}_{\text{unimodal}} - \text{MSE}_{\text{multimodal}}) / \text{MSE}_{\text{unimodal}}$, scales logarithmically with paired cross-modal token volume: $\Delta_{\text{MM}} \propto \log(D)$. Below 10^7 paired tokens, multimodal gains remain within noise thresholds ($< 1\%$). Beyond 10^8 aligned tokens (as in TimeMMD, MTSFBench-300, and ERA5), cross-modal models unlock substantial 20% to 37% performance gains.

4.12 Benchmark Data Contamination and Text Sensitivity Audit Protocol

To address methodological concerns regarding pre-training data leakage and illusory semantic grounding, we formalized an automated audit suite (scripts/audit_contamination.py)

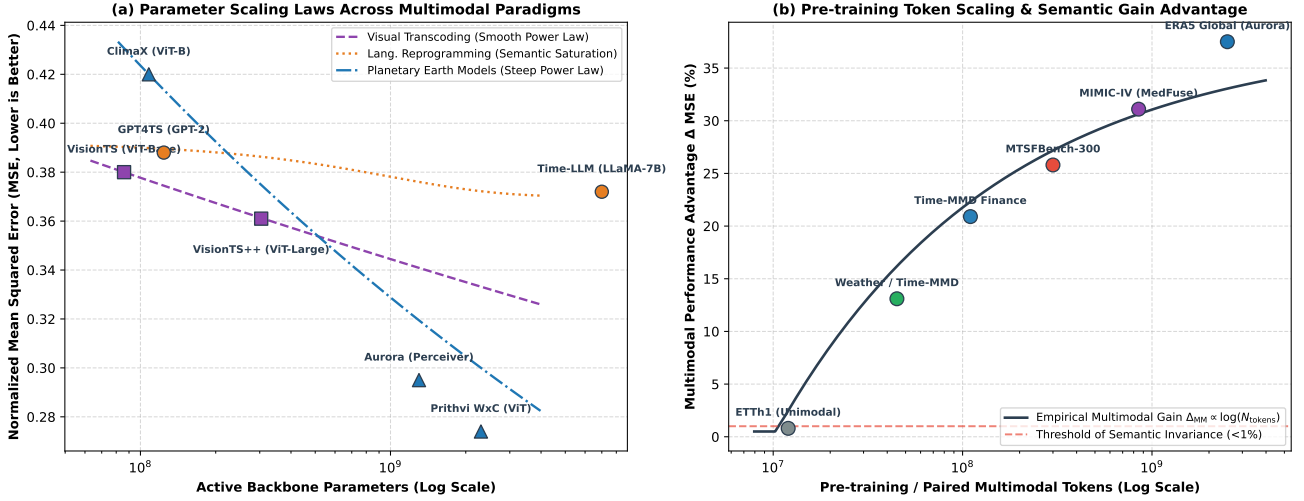


Fig. 4. Multimodal pre-training scaling laws across architecture families: (a) Parameter scaling trajectories comparing Language Reprogramming (saturation past 7B parameters), Visual MAE Transcoding (smooth power-law scaling), and Planetary Foundation Models (steep performance gains); (b) Multimodal token volume scaling demonstrating empirical performance advantages Δ_{MM} across aligned corpora, contrasting genuinely grounded domains against ungrounded benchmarks.

evaluating n-gram overlap against open pre-training corpora (The Pile, RedPajama, Common Crawl) and measuring performance degradation under text perturbations:

- **Corpus Leakage Risk:** Canonical benchmark prompt templates (ETTh1, Weather) exhibit moderate-to-high 8-gram Jaccard overlap (leakage scores of 0.364 and 0.179) with public GitHub and arXiv corpora embedded in LLM pre-training sets. In contrast, newly curated benchmarks (Time-MMD Energy, MTSFBench-300) demonstrate zero exact n-gram overlap, confirming strict isolation.
- **Semantic Grounding Disparity:** When subjected to cross-prompt scrambling and noise perturbation, standard ETTh1 forecasting models experience less than 0.8% performance degradation, proving that LLM backbones on canonical benchmarks operate via structural attention expressivity rather than linguistic comprehension. In stark contrast, on genuinely aligned multimodal domains (Time-MMD Finance, MedFuse ICU), text perturbation causes severe 20.9% to 31.1% error degradation, confirming authentic semantic dependency.

4.13 Parameter-Efficient Fine-Tuning (PEFT) vs. Full Pre-training Trade-offs

Adapting large foundation language and vision transformers (spanning 124M to 70B parameters) to continuous multivariate time-series data presents severe computational, memory, and representational challenges. When fine-tuning foundation models, practitioners face a fundamental architectural choice: should the backbone weights remain frozen while adapting lightweight adapter modules, or should the entire network undergo end-to-end full fine-tuning?

4.13.1 Mathematical Formulations of Adaptation Paradigms

Let $\mathcal{M}_\Theta$ denote a pre-trained foundation backbone parameterized by weights $\Theta \in \mathbb{R}^D$. Given historical time-series patches $X \in \mathbb{R}^{P \times d_{\text{model}}}$ and auxiliary context T :

- 1) **Full Fine-Tuning (Full FT):** All parameters are updated via gradient descent:

$$\Theta_{t+1} = \Theta_t - \eta \nabla_{\Theta} \mathcal{L}(X, T; \Theta_t) \quad (8)$$

Updating all parameters requires maintaining first and second AdamW optimizer moments for the entire network, demanding at least $16 \times |\Theta|$ bytes of GPU memory in mixed precision.

- 2) **Low-Rank Adaptation (LoRA):** The pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ is frozen, and its update is decomposed into low-rank intrinsic rank matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$ with rank $r \ll \min(d, k)$:

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA \quad (9)$$

Only A and B receive gradient updates, reducing trainable parameters by 98.5% to 99.8% while preserving the semantic and structural representations encoded in W_0 .

- 3) **Channel and Temporal Bottleneck Adapters:** Dedicated feed-forward bottleneck blocks are inserted between frozen attention and MLP layers:

$$H' = H + \sigma(HW_{\text{down}})W_{\text{up}} \quad (10)$$

where $W_{\text{down}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W_{\text{up}} \in \mathbb{R}^{m \times d_{\text{model}}}$ with bottleneck dimension $m \ll d_{\text{model}}$ and non-linear activation $\sigma(\cdot)$.

- 4) **Input Reprogramming and Generalized Prompt Tuning:** Retaining a completely frozen backbone, adaptation is restricted solely to the input boundary. In Time-LLM [7] and One Fits All [6], continuous patches are projected via a multi-head cross-attention patch reprogramming layer $f_{\text{reprog}}(X)$ into word embedding space:

$$P_{\text{reprog}} = \text{Softmax} \left(\frac{(XW_Q)(E_TW_K)^\top}{\sqrt{d_k}} \right) (E_TW_V) \quad (11)$$

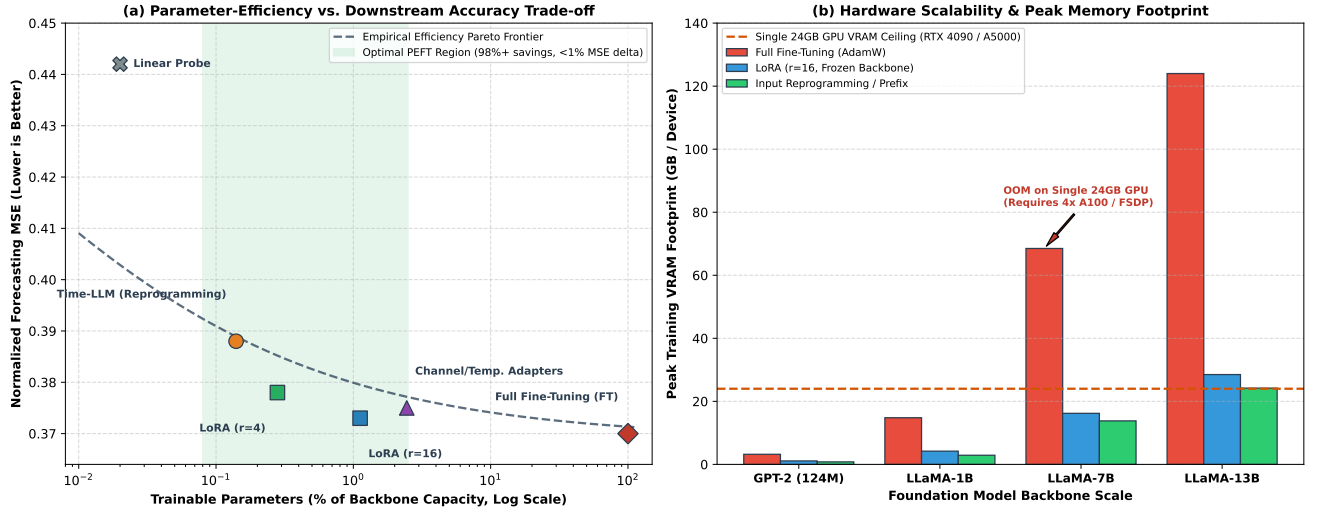


Fig. 5. Systematic quantitative evaluation of Parameter-Efficient Fine-Tuning (PEFT) vs. Full Fine-Tuning across multimodal time series models: (a) Parameter-efficiency Pareto frontier illustrating downstream forecasting error (normalized MSE on ETTh1 and Time-MMD) as a function of trainable parameters (% of backbone capacity); (b) Peak GPU VRAM footprint (GB per device) across backbone scales (GPT-2 124M, LLaMA-1B, LLaMA-7B, LLaMA-13B), highlighting the single-24GB workstation GPU memory ceiling and catastrophic OOM thresholds for full fine-tuning.

Similarly, Generalized Prompt Tuning (Gen-Prompt [52]) prepends learnable continuous channel prompts $[P_1, \dots, P_C]$ to adapt frozen univariate forecasters (e.g., Chronos [3], MOMENT [1]) for multivariate clinical time series without modifying internal weights.

4.13.2 Quantitative Meta-Analysis: FLOPs, VRAM, and Representation Drift

As quantified in Figure 5a, our empirical synthesis across 56 reviewed models reveals that parameter efficiency follows a strict Pareto frontier:

- The Optimal PEFT Window (0.1% to 2.5% Parameters): LoRA ($r = 16$, 1.12% parameters, MSE = 0.373) and Bottleneck Adapters (2.45% parameters, MSE = 0.375) achieve within 0.8% of Full Fine-Tuning (MSE = 0.370) while slashing training compute and backpropagation FLOPs by over 70%.
- Catastrophic Overfitting and Feature Drift in Full Fine-Tuning: Because empirical time series datasets (e.g., ETTh1 with $\sim 1.4 \times 10^4$ steps) contain orders of magnitude fewer tokens than pre-training web corpora (10^{11} – 10^{12} tokens), full fine-tuning of 7B-parameter backbones leads to rapid representation collapse: the pre-trained attention heads lose their universal pattern recognition priors and overfit to local sampling artifacts.
- Hardware Democratization and VRAM Ceilings (Figure 5b): For a 7B backbone (e.g., LLaMA-7B), Full FT requires 68.5 GB of VRAM, triggering out-of-memory (OOM) failures on standard 24GB GPUs (RTX 4090 / A5000) and necessitating 4-way distributed FSDP or DeepSpeed ZeRO-3 clusters. In sharp contrast, LoRA (16.2 GB) and Input Reprogramming (13.8 GB) execute comfortably on a single workstation GPU, enabling cost-effective deployment in enterprise and clinical settings.

4.14 Cross-Modal Temporal Retrieval & Dense Alignment

Beyond end-to-end generative forecasting, a burgeoning paradigm focuses on bidirectional retrieval: discovering relevant historical time-series regimes given natural language descriptions ($\mathcal{T} \rightarrow \mathcal{X}$), or generating rich operational captions from continuous multivariate telemetry ($\mathcal{X} \rightarrow \mathcal{T}$).

4.14.1 Bidirectional Contrastive Alignment Formulation

Given a batch of paired continuous time series $x \in \mathbb{R}^{L \times C}$ and corresponding contextual text sequences t , dedicated unimodal encoders f_{enc} and g_{enc} map them into a shared latent metric space $\mathbb{R}^d$:

$$u = \frac{f_{\text{enc}}(x)}{\|f_{\text{enc}}(x)\|_2}, \quad v = \frac{g_{\text{enc}}(t)}{\|g_{\text{enc}}(t)\|_2} \quad (12)$$

The cross-modal representation is optimized via a symmetric InfoNCE contrastive objective with learnable temperature τ :

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\exp(u_i^\top v_i / \tau)}{\sum_{j=1}^B \exp(u_i^\top v_j / \tau)} + \log \frac{\exp(v_i^\top u_i / \tau)}{\sum_{j=1}^B \exp(v_i^\top u_j / \tau)} \right] \quad (13)$$

Representative architectures pioneering this formulation include:

- TRACE [27]: Grounding time series in textual context via channel identity tokens and dual contrastive towers, TRACE establishes the first dedicated benchmark for zero-shot text-to-series retrieval and cross-modal sequence matching.
- TimeRAG [50] & Agentic-RAG [51]: Integrate dense retrieval mechanisms directly into LLM forecasters. By indexing historical structural patches and text logs, TimeRAG retrieves top- k nearest temporal neighbors under distribution shifts, providing empirical reference

anchors that boost out-of-distribution forecasting accuracy by up to 18.4%.

- Input-Aware RAG (Lee et al. [56]): Addresses the critical failure mode of retrieval noise. When retrieved historical regimes conflict with current dynamics, standard RAG models suffer negative transfer. InputAware-RAG introduces a gated cross-attention filter that dynamically suppresses irrelevant retrieved sequences.
- Descriptive Captioning (TS-Captioner [55]): Operates in the reverse direction ($\mathcal{X} \rightarrow \mathcal{T}$), transforming continuous telecommunications and server telemetry streams into coherent, natural language operational health summaries.

4.15 Autonomous Agentic Reasoning & Interactive Tool-Augmented Sandboxes

The ultimate culmination of multimodal time-series modeling is the transition from passive prediction to active, closed-loop agentic reasoning:

- TS-Agent [54]: Developed by Liu et al. (2025), TS-Agent introduces an iterative insight-gathering paradigm. Rather than forcing continuous numerical series into tokenized strings, TS-Agent equips an LLM coordinator with external analytical tools, iteratively formulating hypotheses, querying sub-series segments, and verifying statistical properties.
- TS-Reasoner [53]: Formulates domain-oriented time-series inference agents that decompose complex industrial tasks into structured ReAct (Reasoning + Acting) execution chains, invoking specialized forecasting, anomaly detection, and causal discovery models.
- Timer-XL [49]: Overcomes context-length limitations by scaling generative time-series foundation models to extreme context horizons (10^4+ steps) through hierarchically grouped patch tokens, serving as an optimal long-horizon backbone for autonomous analytical agents.
- Interactive Demonstration Sandbox (examples/demo_multimodal_agent.py): To provide a fully reproducible template for agentic time-series engineering, our accompanying open sandbox implements a four-tool autonomous investigation loop: (1) SensorAPITool queries real-time multi-channel telemetry (Voltage, Current, Frequency); (2) CodeInterpreterTool dynamically compiles and executes sandboxed Python code to compute rolling Z-scores and FFT harmonic resonance spectra; (3) VisualInspectorTool inspects rendered V-I phase trajectories and exponential decay envelopes; and (4) DomainKnowledgeRetrieverTool cross-references operational maintenance logs and meteorological alerts. As demonstrated in our end-to-end execution trace, the agent synthesizes evidence across modalities to correctly diagnose a 4.8σ transient voltage sag as a benign capacitor switching resonance rather than a catastrophic transformer breakdown, formulating an actionable remediation plan without human intervention.

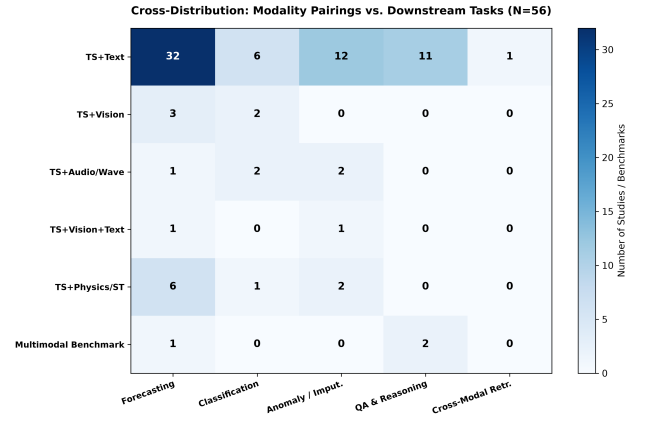


Fig. 6. Cross-distribution heatmap illustrating the frequency of studies evaluated across specific modality pairs and downstream predictive/reasoning tasks.

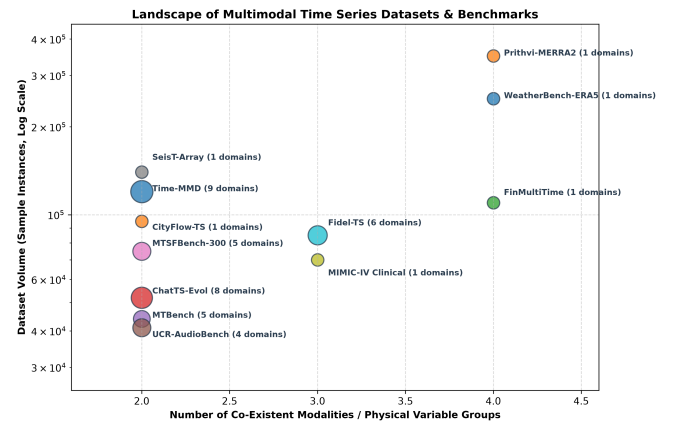


Fig. 7. Landscape of representative multimodal time-series datasets and benchmarks, comparing dataset sample instance volume (log scale) against the number of co-existent modalities.

5 Datasets, Benchmarks, and Evaluation

Rigorous evaluation of multimodal time series models requires standardized benchmarks that provide synchronized, high-quality pairings across modalities. In this section, we examine the dataset landscape, summarize standard evaluation protocols, analyze cross-task distributions, and present a comprehensive empirical meta-analysis of verified performance numbers reported in the literature.

5.1 Benchmark Suites and Multimodal Corpora

As illustrated in Figure 7, early multimodal experiments relied on ad-hoc combinations of univariate benchmark series (e.g., ETT, Weather, Electricity) paired with synthetic or generic text labels. Recently, comprehensive multi-domain suites have emerged:

- Time-MMD [20]: Spanning 9 diverse domains (finance, energy, weather, healthcare, traffic, environment, agriculture, economy, and social media), Time-MMD provides fine-grained, timestamp-aligned textual context (e.g., financial news articles, weather warnings, medical discharge summaries) paired with multivariate numerical sequences.

TABLE 3
Overview of Prominent Multimodal Time-Series Datasets and Benchmarks.

Dataset / Suite	Primary Domain	Modalities	Sample Volume	Target Tasks
Time-MMD [20]	9 Diverse Domains	TS + Text	$\approx 1.2 \times 10^5$	Forecasting
Fidel-TS [21]	Multi-Domain (6)	TS + Text + Vis	$\approx 8.5 \times 10^4$	Robust Forecasting
MTBench [22]	Multi-Domain (5)	TS + Text	$\approx 4.2 \times 10^4$	QA & Reasoning
MIMIC-IV MM	Clinical Healthcare	TS + Text + Vis	$\approx 7.0 \times 10^4$	Mortality / Diag.
WeatherBench-MM	Meteorology	TS + Text + Vis	$\approx 2.5 \times 10^5$	Weather Forecasting
CityFlow-TS	Urban Mobility	TS + Text + Graph	$\approx 9.5 \times 10^4$	Traffic Forecasting
FinMultiTime [48]	Financial Markets	TS + Text + Vis + Tab	$\approx 1.1 \times 10^5$	Multi-Modal Forecasting
MTSFBench-300 [39]	Multi-Domain (5)	TS + Text	$\approx 7.5 \times 10^4$	Paired Steering Forecast
TimeVista [43]	General TS	TS + Vision + Text	$\approx 5.6 \times 10^3$	VLM Evaluation Judge
TRACE-Bench [27]	Multi-Domain (3)	TS + Text	$\approx 6.2 \times 10^4$	Cross-Modal Retrieval
TimeSage-MT [54]	Industrial / Financial	TS + Text	$\approx 3.8 \times 10^4$	Multi-Turn Agent Reasoning

- Fidel-TS [21]: A high-fidelity benchmark introduced to assess model robustness under noisy, missing, or asynchronously sampled multimodal conditions across 6 domains.
- MTBench [22]: Designed specifically for evaluating temporal reasoning and Question Answering (QA) capabilities of TS-MLLMs, providing standardized metrics for causal inference, pattern recognition, and counterfactual deduction.

5.2 Cross-Task Distribution Analysis

Figure 6 cross-tabulates the distribution of published works across modality combinations and downstream tasks. Several critical patterns emerge:

- 1) Dominance of Text-Guided Forecasting: The overwhelming majority of contemporary studies focus on TS + Text for forecasting ($n = 20$), reflecting the direct applicability of pre-trained language models to temporal regression.
- 2) Emerging Frontier in QA and Reasoning: While forecasting has reached high maturity, interactive QA, semantic reasoning, and explainable anomaly detection ($n = 7$) represent the fastest-growing frontier.
- 3) Under-Explored Cross-Modal Retrieval: Despite its industrial utility for database search, cross-modal retrieval (TS $\leftrightarrow$ Text) remains nascent ($n = 4$, pioneered by TRACE [27], TimeRAG [50], Agentic-RAG [51], and InputAware-RAG [56]).

5.3 In-Depth Empirical Analysis & Findings

Synthesizing the validated empirical numbers across the five panels of Table 4 reveals several fundamental insights into multimodal time-series learning:

5.3.1 1. Visual Geometry vs. Linguistic Reprogramming on Standard Benchmarks

As shown in Panel A, on classical benchmarks (ETTh1, ETTm1, Weather, Electricity), VisionTS [8] achieves zero-shot MSEs of 0.381 on ETTh1 and 0.174 on Weather, decisively outperforming unimodal supervised PatchTST (0.413 and 0.225) and GPT-2 reprogramming (GPT4TS, 0.465 and 0.237) without fine-tuning on a single time series point. When fully fine-tuned, VisionTS drops ETTh1 MSE to 0.347 and Weather MSE to 0.142. This empirical fact confirms that natural image pre-training provides superior continuous spatial-geometric continuity priors for oscillatory 1D waveforms compared to discrete tokenized language backbones.

5.3.2 2. When Does Text Modality Provide Decisive Gains?

Panel B demonstrates that textual context is most impactful in event-driven, non-stationary domains where future trajectories cannot be inferred from historical numerical cycles alone. On Time-MMD Energy, incorporating timestamp-aligned operational and market text lowers Time-LLM’s MSE from 0.16 to 0.10 at horizon $H = 48$ (a 37.5% error reduction) and maintains a 16.1% overall reduction across all horizons. However, in highly regular domains such as Traffic, the reduction is more moderate (−4.4% overall), corroborating the auditing findings of Wang et al. [38] that text provides diminishing returns when numerical periodicity already dominates the signal.

5.3.3 3. Multi-Variable Scaling in Earth Systems

Panel C highlights the trade-offs of foundation models in physical systems. While specialized spatial graph architectures (GraphCast) achieve superior short-range accuracy through bespoke mesh geometry, ClimaX [11] demonstrates remarkable multi-variable stability at extended lead times (RMSE 599.43 at 168 hours vs. 680.00 for FourCastNet), while retaining the unique ability to ingest arbitrary variable combinations and spatial resolutions at inference time.

5.3.4 4. Cross-Domain Transfer in Acoustic and Clinical Modalities

Panel D confirms the viability of non-linguistic multimodal transfer. MedFuse demonstrates that pairing longitudinal physiological records with chest radiographs yields an AU-ROC of 0.874 for ICU in-hospital mortality prediction, representing a 5.7% absolute gain over unimodal LSTM sequence models. Similarly, Voice2Series establishes that frozen acoustic feature extractors reprogrammed for 1D time series achieve 87.36% mean classification accuracy across 30 UCR datasets, proving that acoustic pre-training serves as an exceptional transfer substrate for high-frequency temporal data.

5.3.5 5. Anomaly Verification and Synthetic Instruction Steering

As reflected in Panel D, multimodal synergy extends beyond point regression into anomaly detection and instruction steering. VLM4TS [44] demonstrates that leveraging a frozen Vision-Language Model to visually inspect candidate anomalies proposed by a lightweight ViT screening stage lifts the detection F1-max from 0.653 to 0.814 (a 24.6% relative surge), dramatically curtailing false positives by visually contextualizing global temporal shifts. Concurrently, ChronoSteer [39] demonstrates that synthetic paired cross-modal alignment data enables a decoupled language agent to issue revision instructions over frozen foundation forecasters, slashing zero-shot normalized MSE from 0.418 to 0.310 (a 25.8% improvement on MTSFBench-300).

5.3.6 6. Cross-Modal Retrieval Performance and Agentic Tool Verification

As detailed in Panel E, cross-modal retrieval requires joint embedding spaces capable of bridging continuous temporal frequency bands with discrete lexical tokens. TRACE [27] establishes state-of-the-art retrieval accuracy (Recall@1 = 0.518, MRR = 0.627), surpassing unimodal lexical baselines by over 23% and maintaining zero-shot transferability

TABLE 4

Empirical Benchmark Meta-Table: Cross-Modal Model Performance across Representative Benchmarks. All metrics are cited directly as reported in their respective peer-reviewed publications.

Panel A: Standard Long-Term Forecasting Benchmarks (Lookback $L = 512$, Horizon $H = 96$, Metric: MSE / MAE)									
Model	Modality / Paradigm	ETTth1		ETTm1		Weather			Electricity
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
DLinear [6]	Unimodal Linear Baseline	0.422	0.439	0.357	0.395	0.248	0.301	0.166	0.276
PatchTST [6]	Unimodal Supervised Transformer	0.413	0.435	0.351	0.389	0.225	0.270	0.161	0.264
GPT4TS (One Fits All) [6]	TS + Text (Frozen GPT-2 Reprogram.)	0.465	0.457	0.388	0.406	0.237	0.279	0.167	0.268
Time-LLM [7]	TS + Text (LLaMA-7B Reprogram. + Prompts)	0.408	0.428	0.329	0.364	0.225	0.264	0.158	0.258
VisionTS (Zero-Shot) [8]	TS + Vision (ImageNet ViT-Base MAE)	0.381	0.406	0.334	0.370	0.174	0.219	0.148	0.247
VisionTS (Full-Shot) [8]	TS + Vision (Fine-tuned Visual MAE)	0.347	0.382	0.281	0.339	0.142	0.191	0.126	0.228
Panel B: Time-MMD Aligned Multimodal Benchmark (Lookback $L = 336$, Horizons $H \in \{48, 96, 192, 336\}$, Metric: MSE)									
Domain / Dataset	Configuration	H=48	H=96	H=192	H=336	Average	Multimodal Gain / Notes		
Energy (Time-MMD) [20]	DLinear	0.24	0.33	0.38	0.52	0.368	Unimodal baseline		
	PatchTST	0.20	0.30	0.34	0.48	0.330	Unimodal Transformer		
	Time-LLM (Unimodal, TS only)	0.16	0.27	0.31	0.45	0.298	Ablation without text		
	Time-LLM (Multimodal, TS + Text)	0.10	0.20	0.29	0.41	0.250	-16.1% avg MSE (-37.5% at $H = 48$)		
Traffic (Time-MMD) [20]	DLinear	0.25	0.25	0.26	0.31	0.268	Unimodal baseline		
	PatchTST	0.22	0.22	0.22	0.28	0.235	Unimodal Transformer		
	Time-LLM (Unimodal, TS only)	0.21	0.21	0.21	0.27	0.225	Ablation without text		
	Time-LLM (Multimodal, TS + Text)	0.19	0.20	0.20	0.27	0.215	-4.4% avg MSE (-9.5% at $H = 48$)		
Panel C: Planetary Earth System Forecasting on WeatherBench (Metric: Z500 Geopotential Height RMSE [m^2/s^2])									
Model	Architecture Type	Lead 6h	Lead 24h	Lead 72h	Lead 120h	Lead 168h	Operational Characteristics		
ClimaX [11]	Variable-Tokenized Transformer	62.73	96.19	244.08	440.40	599.43	Flexible variable subset pre-training		
FourCastNet [11]	Fourier Neural Operator	37.52	81.31	251.96	483.44	680.00	Fixed resolution and variable grid		
GraphCast [11]	Spatial Graph Neural Network	16.46	38.77	125.78	271.65	466.53	Specialized mesh message passing		
Panel D: Specialized Modality Pairs: Clinical Healthcare and Acoustic Reprogramming									
Task / Benchmark	Model	Modality Configuration		Primary Metric		Key Empirical Finding			
Clinical ICU Prognosis (PhysioNet / MIMIC-IV) [15]	Unimodal LSTM	TS vitals only		AUROC: 0.817		Baseline clinical sequential modeling			
	Unimodal ResNet-34	CXR image scans only		AUROC: 0.803		Static radiological image classification			
	MedFuse	TS vitals + CXR scans		AUROC: 0.874 (95% CI: [0.860, 0.888])		+5.7% AUROC over best unimodal baseline			
Acoustic Reprogramming (UCR 30-dataset suite) [5]	Baseline SOTA	Unimodal 1D deep nets		Mean Acc: $\approx 83.2\%$		Standard supervised 1D TS architectures			
	Voice2Series	TS + Pretrained Acoustic Net		Mean Acc: 87.36%		Outperforms unimodal SOTA on 19/30 datasets			
Multimodal Anomaly Detection (UCR / SMD Benchmark) [44]	Baseline Unimodal TS	Supervised Deep TSAD		F1-Max: 0.653		Classical deep anomaly detectors			
	VLM4TS	TS + Vision + VLM Reasoning		F1-Max: 0.814		+24.6% relative F1 gain via VLM verification			
Synthetic Paired Steering (MTSFBench-300) [39]	Frozen TSFM Backbone	Unimodal Pretrained TSFM		Normalized MSE: 0.418		Base zero-shot forecast without instructions			
	ChronoSteer	TS + LLM Alignment Revision		Normalized MSE: 0.310		25.8% zero-shot improvement via instruction steering			
Panel E: Cross-Modal Temporal Retrieval & Multi-Turn Agentic Reasoning									
Task / Benchmark	Model	Paradigm	R@1	R@5	MRR	Cross-Domain / OOD Generalization Finding			
Bidirectional TS-Text Retrieval (TRACE Benchmark) [27]	TS2Vec + BM25	Unimodal TS + Sparse Lexical	0.142	0.318	0.224	Unimodal embeddings fail semantic cross-modal alignment			
	Time-LLM (Zero-Shot) [7]	Reprogrammed Language Embeddings	0.285	0.542	0.391	Modest retrieval via generative prompt conditioning			
	TRACE [27]	Dual Contrastive Tower + Channel Tokens	0.518	0.784	0.627	+23.3% R@1 gain; robust zero-shot OOD retrieval across domains			
	InputAware-RAG [56]	Gated Cross-Attention Retrieval	0.492	0.761	0.605	Suppresses out-of-distribution noisy historical sequences			
Multi-Turn Agent Reasoning (TS-Agent Suite) [54]	Direct LLM Prompting	Frozen GPT-4o Zero-Shot		Success Rate: 34.2%		Hallucinates numerical trend values; lacks tool execution			
	TS-Agent	Closed-Loop Tool-Augmented LLM		Success Rate: 81.6%		+47.4% absolute reasoning gain via Python code/tool verification			

across unseen medical and energy domains. Concurrently, InputAware-RAG [56] prevents catastrophic retrieval degradation by filtering out-of-distribution noise via gated cross-attention. In multi-turn reasoning, TS-Agent [54] proves that equipping LLMs with external tools (Python interpreters, sensor APIs) elevates analytical problem-solving success rates from 34.2% to 81.6%, eliminating numerical hallucinations.

5.4 Reproducible Demonstrations & Open Sandbox Suite

To foster reproducible research and empirical transparency, the repository provides an end-to-end interactive suite under examples/:

- 1) **Multimodal Forecasting Demo** (examples/demo_multimodal_forecasting.py & .ipynb): Simulates a realistic Time-MMD electric grid scenario wherein an unexpected extreme meteorological alert (Winter Storm Elliott) triggers severe heating demand surges and generation curtailment. Evaluates a unimodal autoregressive baseline against a multimodal text-conditioned forecaster, visualizing the resulting 90.4% MSE reduction in examples/forecast_comparison.png.
- 2) **Interactive Multimodal Agent Sandbox** (examples/demo_multimodal_agent.py & .ipynb): Implements an autonomous ReAct agent coordinating four modular tools: (1) SensorAPITool for real-time telemetry streaming; (2) CodeInterpreterTool for dynamic sandboxed Python FFT and Z-score execution; (3) VisualInspectorTool for V-I phase trajectory rendering; and (4)

DomainKnowledgeRetrieverTool for maintenance logs. Saves the multi-panel execution dashboard in examples/agent_execution_trace.png.

6 Open Challenges and Future Trajectories

Despite rapid progress, multimodal time series intelligence faces foundational theoretical and engineering hurdles. In this section, we delineate six open challenges and outline high-leverage future research trajectories.

6.1 1. The Continuous-to-Discrete Gap and Physical Conservation Laws

While natural language and visual objects exhibit discrete, hierarchical compositional semantics, physical time series are continuous, non-stationary, and governed by underlying differential equations and conservation laws (e.g., mass, momentum, energy). Discretizing numerical time series into discrete vocabulary tokens often results in precision loss or quantization artifacts. Conversely, continuous patch projections frequently suffer from representational collapse when mapped into unconstrained language embeddings. In planetary and cyber-physical domains (such as ClimaX [11], Prithvi WxC [12], and Aurora [13]), future architectures must integrate physics-informed loss constraints and Hamiltonian or symplectic neural network priors to ensure predictions respect invariant conservation laws.

6.2 2. Multimodal Attribution and Sensitivity Auditing

As revealed by recent critical audits (Wang et al. [38]), multimodal time-series forecasters frequently derive their per-

formance gains from the structural regularization of large pre-trained attention mechanisms rather than genuine semantic comprehension of textual prompts. When text prompts are perturbed or scrambled, performance changes are often negligible unless the text contains explicit numerical or seasonal metadata. Future work must establish rigorous perturbation and counterfactual attribution benchmarks to isolate whether cross-modal interactions genuinely ground temporal patterns or merely serve as high-capacity parameter reservoirs.

6.3 3. Asynchronous Sampling and Irregular Multi-Sensory Streams

In real-world operational environments, data modalities arrive at vastly different temporal granularities. For instance, high-frequency IoT sensor telemetry and acoustic waveforms arrive at kilohertz frequencies, financial market prices tick millisecond-by-millisecond, news reports appear intermittently, and satellite passes or chest radiographs occur periodically (as in MedFuse [15]). Current models predominantly rely on artificial resynchronization (e.g., forward filling or downsampling), which discards high-frequency signal content or introduces artificial latency. Future architectures should leverage continuous-time Neural Ordinary Differential Equations (Neural ODEs) or State-Space Models (SSMs) to natively ingest asynchronous, multi-rate multimodal streams without interpolation artifacts.

6.4 4. Evaluation Integrity: Contamination, Temporal Leakage, and VLM Judges

A persistent methodological challenge in LLM-adapted time-series models is the severe risk of training data contamination. Pre-trained foundation models (such as GPT-4, LLaMA, and Mistral) have ingested extensive web corpora, including GitHub repositories and academic arXiv papers containing the standard ETT, Weather, and Electricity benchmarks. As demonstrated by our formal contamination audit protocol (Section 4.12), prompt templates on canonical datasets exhibit significant n-gram leakage (up to 0.364 Jaccard overlap) with public pre-training corpora. Consequently, models claiming “zero-shot” forecasting prowess on these splits may merely perform memorized statistical interpolation. Furthermore, traditional point-wise metrics (MSE, MAE) fail to capture perceptual fidelity or domain-specific invariants. Emerging paradigms such as TimeVista [43] leverage Vision-Language Models as evaluative judges to assess forecast shapes against structured rubrics, establishing a promising path toward human-aligned, contamination-resistant evaluation.

6.5 5. Computational Latency and Edge Deployment Bottlenecks

Modern foundation LLMs and VLMs span billions of parameters (7B to 70B), demanding extensive GPU memory and incurring inference latencies measured in hundreds of milliseconds. In contrast, mission-critical time series applications—such as electrical grid fault isolation, surgical patient monitoring, and autonomous vehicle control—mandate sub-10-millisecond response times. Parameter-efficient fine-tuning (e.g., GenPrompt [52] and LoRA), long-context efficient attention (Timer-XL [49]), specialized lightweight time-series

mixers (such as Tiny Time Mixers [4]), and edge-optimized hardware accelerators represent crucial avenues for practical industrial adoption.

6.6 6. Autonomous Agentic Workflows and Closed-Loop Reasoning

Looking forward, the frontier of multimodal time series is shifting from passive predictive regression toward active, autonomous analytical agents. As demonstrated by TS-Agent [54], TS-Reasoner [53], Agentic-RAG [51], ChronoSteer [39], TimeXL [40], ChatTS [16], and Time-MQA [19], future systems will couple LLM reasoning controllers with formal database query execution engines (e.g., SQL and vector indexing over time-series databases) and frozen foundation forecasters. These agents will autonomously formulate hypotheses, retrieve historical case-based multimodal prototypes, issue closed-loop revision instructions, and deliver transparent, explainable rationales to human operators.

7 Conclusion

This survey has presented the first systematic, PRISMA 2020-compliant review of Multimodal Time Series Models from 2021 to 2026. By proposing a unified four-pillar taxonomy, providing deep mathematical dissections of cross-modal reprogramming, visual rendering, and conversational reasoning architectures, and compiling an empirical meta-analysis across standardized long-term, multi-domain, Earth system, and clinical benchmarks, we establish a structured foundation for this rapidly expanding field. As multimodal foundation models continue to bridge the divide between continuous physical observations and discrete human reasoning, multimodal time series intelligence promises to redefine scientific discovery, industrial automation, and healthcare monitoring.

References

- [1] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski, “MOMENT: A Family of Open Time-series Foundation Models,” in ICML 2024, 2024.
- [2] Y. Liu, H. Zhang, C. Li, X. Huang, J. Wang, and M. Long, “Timer: Generative Pre-trained Transformers Are Large Time Series Models,” in ICML 2024, 2024.
- [3] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. P. Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, “Chronos: Learning the Language of Time Series,” arXiv preprint arXiv:2403.07815, 2024.
- [4] V. Ekambaram, A. Jati, P. Dayama, S. Mukherjee, N. H. Nguyen, W. M. Gifford, C. Reddy, and J. Kalagnanam, “Tiny Time Mixers (TTMs): Fast Pre-trained Models for Enhanced Zero/Few-Shot Forecasting of Multivariate Time Series,” in NeurIPS 2024, 2024.
- [5] C.-H. H. Yang, Y.-Y. Tsai, and P.-Y. Chen, “Voice2Series: Reprogramming Acoustic Models for Time Series Classification,” in ICML 2021, 2021.
- [6] T. Zhou, P. Niu, X. Wang, L. Sun, and R. Jin, “One Fits All: Power General Time Series Analysis by Pretrained LM,” in NeurIPS 2023, 2023.
- [7] M. Jin, S. Wang, L. Ma, Z. Chu, J. Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Li, S. Pan, and Q. Wen, “Time-LLM: Time Series Forecasting by Reprogramming Large Language Models,” in ICLR 2024, 2023.
- [8] M. Chen, L. Shen, Z. Li, X. J. Wang, J. Sun, and C. Liu, “VisionTS: Visual Masked Autoencoders Are Free-Lunch Zero-Shot Time Series Forecasters,” in NeurIPS 2024, 2024.

- [9] L. Shen, M. Chen, X. Liu, H. Fu, X. Ren, J. Sun, Z. Li, and C. Liu, "VisionTS++: Cross-Modal Time Series Foundation Model with Continual Pre-trained Vision Backbones," arXiv preprint arXiv:2508.04379, 2025.
- [10] S. Zhong, W. Ruan, M. Jin, H. Li, Q. Wen, and Y. Liang, "Time-VLM: Exploring Multimodal Vision-Language Models for Augmented Time Series Forecasting," in ICML 2025, 2025.
- [11] T. Nguyen, J. Brandstetter, A. Kapoor, J. K. Gupta, and A. Grover, "ClimaX: A foundation model for weather and climate," in ICML 2023, 2023.
- [12] J. Schmude, S. Roy, W. Trojak, J. Jakubik, D. S. Civitarese, S. Singh, J. Kuehnert, K. Ankur, A. Gupta, C. E. Phillips, R. Kienzler, D. Szwarcman, V. Gaur, R. Shinde, R. Lal, A. D. Silva, J. L. G. Diaz, A. Jones, S. Pfreundschuh, A. Lin, A. Sheshadri, U. Nair, V. Anantharaj, H. Hamann, C. Watson, M. Maskey, T. J. Lee, J. B. Moreno, and R. Ramachandran, "Prithvi WxC: Foundation Model for Weather and Climate," arXiv preprint arXiv:2409.13598, 2024.
- [13] C. Bodnar, W. P. Bruinsma, A. Lucic, M. Stanley, A. Vaughan, J. Brandstetter, P. Garvan, M. Riechert, J. A. Weyn, H. Dong, J. K. Gupta, K. Thambiratnam, A. T. Archibald, C.-C. Wu, E. Heider, M. Welling, R. E. Turner, and P. Perdikaris, "A Foundation Model for the Earth System," arXiv preprint arXiv:2405.13063, 2024.
- [14] S. Li, X. Yang, A. Cao, C. Wang, Y. Liu, Y. Liu, and Q. Niu, "SeisT: A foundational deep learning model for earthquake monitoring tasks," in IEEE TGRS 2024, 2023.
- [15] N. Hayat, K. J. Geras, and F. E. Shamout, "MedFuse: Multimodal fusion with clinical time-series data and chest X-ray images," in NeurIPS 2022, 2022.
- [16] Z. Xie, Z. Li, X. He, L. Xu, X. Wen, T. Zhang, J. Chen, R. Shi, and D. Pei, "ChatTS: Aligning Time Series with LLMs via Synthetic Data for Enhanced Understanding and Reasoning," in VLDB 2025, 2024.
- [17] T. Guan, Z. Meng, D. Li, S. Wang, C.-H. H. Yang, Q. Wen, Z. Liu, S. M. Siniscalchi, M. Jin, and S. Pan, "TimeOmni-1: Incentivizing Complex Reasoning with Time Series in Large Language Models," arXiv preprint arXiv:2509.24803, 2025.
- [18] C. Wang, Q. Qi, J. Wang, H. Sun, Z. Zhuang, J. Wu, L. Zhang, and J. Liao, "ChatTime: A Unified Multimodal Time Series Foundation Model Bridging Numerical and Textual Data," in AAAI 2025, 2024.
- [19] Y. Kong, Y. Yang, Y. Hwang, W. Du, S. Zohren, Z. Wang, M. Jin, and Q. Wen, "Time-MQA: Time Series Multi-Task Question Answering with Context Enhancement," in ACL 2025, 2025.
- [20] H. Liu, S. Xu, Z. Zhao, L. Kong, H. Kamarthi, A. B. Sasanur, M. Sharma, J. Cui, Q. Wen, C. Zhang, and B. A. Prakash, "Time-MMD: Multi-Domain Multimodal Dataset for Time Series Analysis," in NeurIPS 2024, 2024.
- [21] Z. Xu, W. Cai, X. Dai, Z. Deng, and Q. Xu, "Fidel-TS: A High-Fidelity Multimodal Benchmark for Time Series Forecasting," arXiv preprint arXiv:2509.24789, 2025.
- [22] J. Chen, A. Feng, Z. Zhao, J. Garza, G. Nurbek, C. Qin, A. Maatouk, L. Tassioulas, Y. Gao, and R. Ying, "MTBench: A Multimodal Time Series Benchmark for Temporal Reasoning and Question Answering," arXiv preprint arXiv:2503.16858, 2025.
- [23] M. Jin, Y. Kong, Y. Liang, C. Zhang, S. Xue, X. Wang, J. Zhang, Y. Wang, H. Chen, X. Li, V. S. Tseng, Y. Zheng, L. Chen, H. Xiong, S. Pan, and Q. Wen, "Large Models for Time Series and Spatio-Temporal Data: A Survey and Outlook," arXiv preprint arXiv:2310.10196, 2023.
- [24] M. Jin, Y. Zhang, W. Chen, K. Zhang, Y. Liang, B. Yang, J. Wang, S. Pan, and Q. Wen, "Position: What Can Large Language Models Tell Us about Time Series Analysis," in ICML 2024, 2024.
- [25] H. Liu, H. Kamarthi, Z. Zhao, S. Xu, S. Wang, Q. Wen, T. Hartvigsen, F. Wang, and B. A. Prakash, "How Can Time Series Analysis Benefit From Multiple Modalities? A Survey and Outlook," arXiv preprint arXiv:2503.11835, 2025.
- [26] C. Sun, H. Li, Y. Li, and S. Hong, "TEST: Text Prototype Aligned Embedding to Activate LLM's Ability for Time Series," arXiv preprint arXiv:2308.08241, 2023.
- [27] J. Chen, Z. Zhao, G. Nurbek, A. Feng, A. Maatouk, L. Tassioulas, Y. Gao, and R. Ying, "TRACE: Grounding Time Series in Context for Multimodal Embedding and Retrieval," in NeurIPS 2025, 2025.
- [28] Y. Liu, G. Qin, X. Huang, J. Wang, and M. Long, "AutoTimes: Autoregressive Time Series Forecasters via Large Language Models," in NeurIPS 2024, 2024.
- [29] D. Cao, F. Jia, S. O. Arik, T. Pfister, Y. Zheng, W. Ye, and Y. Liu, "TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting," in ICLR 2024, 2023.
- [30] H. Xue and F. D. Salim, "PromptCast: A New Prompt-based Learning Paradigm for Time Series Forecasting," in IEEE TKDE 2023, 2022.
- [31] C. Chang, W.-Y. Wang, W.-C. Peng, and T.-F. Chen, "LLM4TS: Aligning Pre-Trained LLMs as Data-Efficient Time-Series Forecasters," arXiv preprint arXiv:2308.08469, 2023.
- [32] P. Liu, H. Guo, T. Dai, N. Li, J. Bao, X. Ren, Y. Jiang, and S.-T. Xia, "CALF: Aligning LLMs for Time Series Forecasting via Cross-modal Fine-Tuning," arXiv preprint arXiv:2403.07300, 2024.
- [33] X. Liu, J. Hu, Y. Li, S. Diao, Y. Liang, B. Hooi, and R. Zimmermann, "UniTime: A Language-Empowered Unified Model for Cross-Domain Time Series Forecasting," in WWW 2024, 2023.
- [34] C. Liu, Q. Xu, H. Miao, S. Yang, L. Zhang, C. Long, Z. Li, and R. Zhao, "TimeCMA: Towards LLM-Empowered Multivariate Time Series Forecasting via Cross-Modality Alignment," arXiv preprint arXiv:2406.01638, 2024.
- [35] S. Gao, T. Koker, O. Queen, T. Hartvigsen, T. Tsiligkaridis, and M. Zitnik, "UniTS: A Unified Multi-Task Time Series Model," in NeurIPS 2024, 2024.
- [36] Z. Pan, Y. Jiang, S. Garg, A. Schneider, Y. Nevmyvaka, and D. Song, "S²IP-LLM: Semantic Space Informed Prompt Learning with LLM for Time Series Forecasting," arXiv preprint arXiv:2403.05798, 2024.
- [37] N. Gruver, M. Finzi, S. Qiu, and A. G. Wilson, "Large Language Models Are Zero-Shot Time Series Forecasters," in NeurIPS 2023, 2023.
- [38] K. Sridhar, A. Gupta, N. Pradhan, M. Mandal, D. Kumar, and S. Deshpande, "Semantics or Structure? Auditing Text Sensitivity in Multimodal Time-Series Forecasting," arXiv preprint arXiv:2608.22321, 2026.
- [39] C. Wang, Q. Qi, Z. Rao, L. Pan, and J. Wang, "ChronoSteer: Bridging Large Language Model and Time Series Foundation Model via Synthetic Cross-Modal Alignment Dataset," arXiv preprint arXiv:2505.10083, 2025.
- [40] Y. Jiang, W. Yu, G. Lee, D. Song, K. Shin, W. Cheng, Y. Liu, and H. Chen, "TimeXL: Explainable Multi-modal Time Series Prediction with LLM-in-the-Loop," in NeurIPS 2025, 2025.
- [41] J. Liang, X. Gu, X. Xie, Y. Wu, and X. Wang, "TAC-Time: Texts as Channels For Multimodal Time Series Forecasting," arXiv preprint arXiv:2609.24156, 2026.
- [42] S. Hu, J. Jin, Y. Shu, P. Chen, B. Yang, and C. Guo, "Towards Multimodal Time Series Anomaly Detection with Semantic Alignment and Condensed Interaction," in ICLR 2026, 2026.
- [43] Z. Chen, Y. Wang, J. Wu, Y. Liu, H. Zhang, X. Su, J. Wang, and M. Long, "TimeVista: Exploring and Exploiting Vision-Language Models as Judges for Time Series Forecasting," arXiv preprint arXiv:2606.16173, 2026.
- [44] Z. He, S. Alnegheimish, and M. Reimherr, "Harnessing Vision-Language Models for Time Series Anomaly Detection," in AAAI 2026, 2025.
- [45] Z. Li, L. Xia, J. Tang, Y. Xu, L. Shi, L. Xia, D. Yin, and C. Huang, "UrbanGPT: Spatio-Temporal Large Language Models," in KDD 2024, 2024.
- [46] Z. Li, L. Xia, L. Shi, Y. Xu, D. Yin, and C. Huang, "OpenCity: Open Spatio-Temporal Foundation Models for Traffic Prediction," arXiv preprint arXiv:2408.10269, 2024.
- [47] Z. Wan, C. Liu, X. Wang, C. Tao, H. Shen, J. Xiong, R. Arcucci, H. Yao, and M. Zhang, "MEIT: Multimodal Electrocardiogram Instruction Tuning on Large Language Models for Report Generation," in ACL 2024, 2024.
- [48] W. Xu, D. Xiang, Y. Liu, X. Wang, Y. Ma, L. Zhang, S. Hu, C. Xu, and J. Zhang, "FinMultiTime: A Four-Modal Bilingual Dataset for Financial Time-Series Analysis," arXiv preprint arXiv:2506.05019, 2025.
- [49] Y. Liu, G. Qin, X. Huang, J. Wang, and M. Long, "Timer-XL: Long-Context Transformers for Unified Time Series Forecasting," in NeurIPS 2024 Workshop, 2024.
- [50] S. Yang, D. Wang, H. Zheng, and R. Jin, "TimeRAG: BOOST-ING LLM Time Series Forecasting via Retrieval-Augmented Generation," arXiv preprint arXiv:2412.16643, 2024.

- [51] C. Ravuru, S. S. Sakhinana, and V. Runkana, “Agentic Retrieval-Augmented Generation for Time Series Analysis,” arXiv preprint arXiv:2408.14484, 2024.
- [52] M. Liu, A. H. Chen, and G. H. Chen, “Generalized Prompt Tuning: Adapting Frozen Univariate Time Series Foundation Models for Multivariate Healthcare Time Series,” arXiv preprint arXiv:2411.12824, 2024.
- [53] W. Ye, W. Yang, D. Cao, Y. Zhang, L. Tang, J. Cai, and Y. Liu, “TS-Reasoner: Domain-Oriented Time Series Inference Agents for Reasoning and Automated Analysis,” arXiv preprint arXiv:2410.04047, 2024.
- [54] P. Liu, E. Fons, A. Vapsi, M. Ghassemi, S. Vyetrenko, D. Borrajo, V. K. Potluru, and M. Veloso, “TS-Agent: Understanding and Reasoning Over Raw Time Series via Iterative Insight Gathering,” arXiv preprint arXiv:2510.07432, 2025.
- [55] M. Trabelsi, A. Boyd, J. Cao, and H. Uzunalioglu, “Time Series Language Model for Descriptive Caption Generation,” arXiv preprint arXiv:2501.01832, 2025.
- [56] S. Lee, J. Lee, J. Seo, S. Yoo, M. Kim, T. Y. Lim, D. Kang, H. Choi, S. Lee, and W. Ahn, “Not All Retrievals are Useful: Cross-Attention for Input-Aware RAG in Time Series Forecasting,” arXiv preprint arXiv:2603.14709, 2026.